



Deep Learning

Dr. Mehran Safayani

safayani@iut.ac.ir

safayani.iut.ac.ir



<https://www.aparat.com/mehran.safayani>

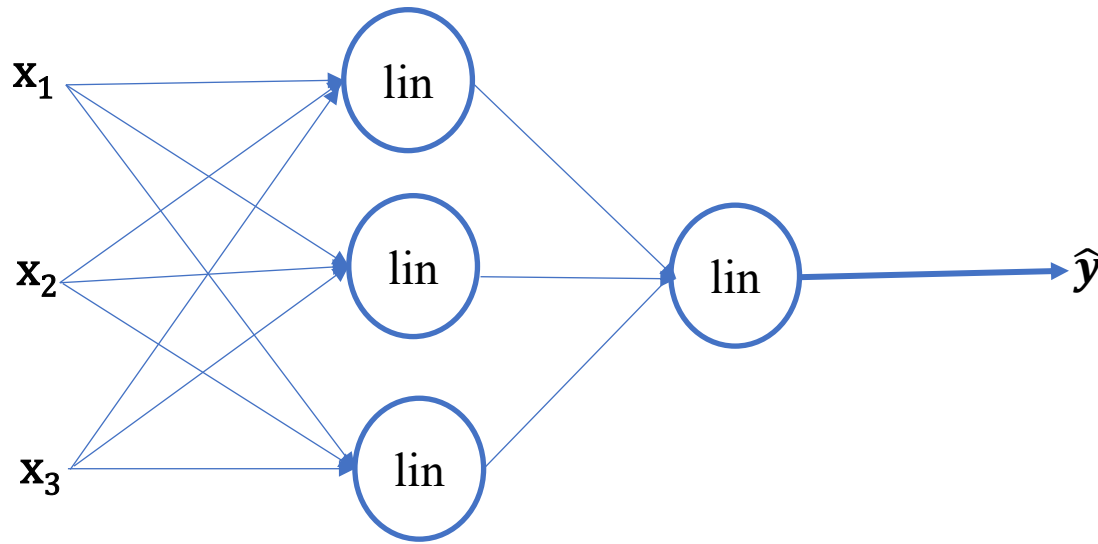


https://github.com/safayani/deep_learning_course



Activation function

Non-linear activation function



• چرا به توابع فعال ساز غیر خطی نیاز داریم؟

• با فرض تابع فعال ساز خطی

- $z^{[1]} = w^{[1]}x + b^{[1]}$
- $a^{[1]} = g(z^{[1]}) = z^{[1]}$
- $z^{[2]} = w^{[2]}a^{[1]} + b^{[2]}$
- $a^{[2]} = g(z^{[2]}) = z^{[2]}$

$$\begin{aligned} a^{[1]} &= z^{[1]} = w^{[1]}x + b^{[1]} \\ a^{[2]} &= z^{[2]} = w^{[2]}a^{[1]} + b^{[2]} \\ &= w^{[2]}(w^{[1]}x + b^{[1]}) + b^{[2]} \\ &= \underbrace{w^{[2]}w^{[1]}}_{w'}x + \underbrace{w^{[2]}b^{[1]} + b^{[2]}}_{b'} \\ &= w'x + b' \end{aligned}$$

Derivatives of activation functions

- Sigmoid:

- $g(z) = \frac{1}{1+e^{-z}} \quad g'(z) = \frac{dg(z)}{dz} = \frac{1}{1+e^{-z}} \left(1 - \frac{1}{1+e^{-z}}\right) = g(z)(1 - g(z))$

- Tanh:

- $g(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}} \quad g'(z) = 1 - \tanh^2(z) = 1 - g^2(z)$

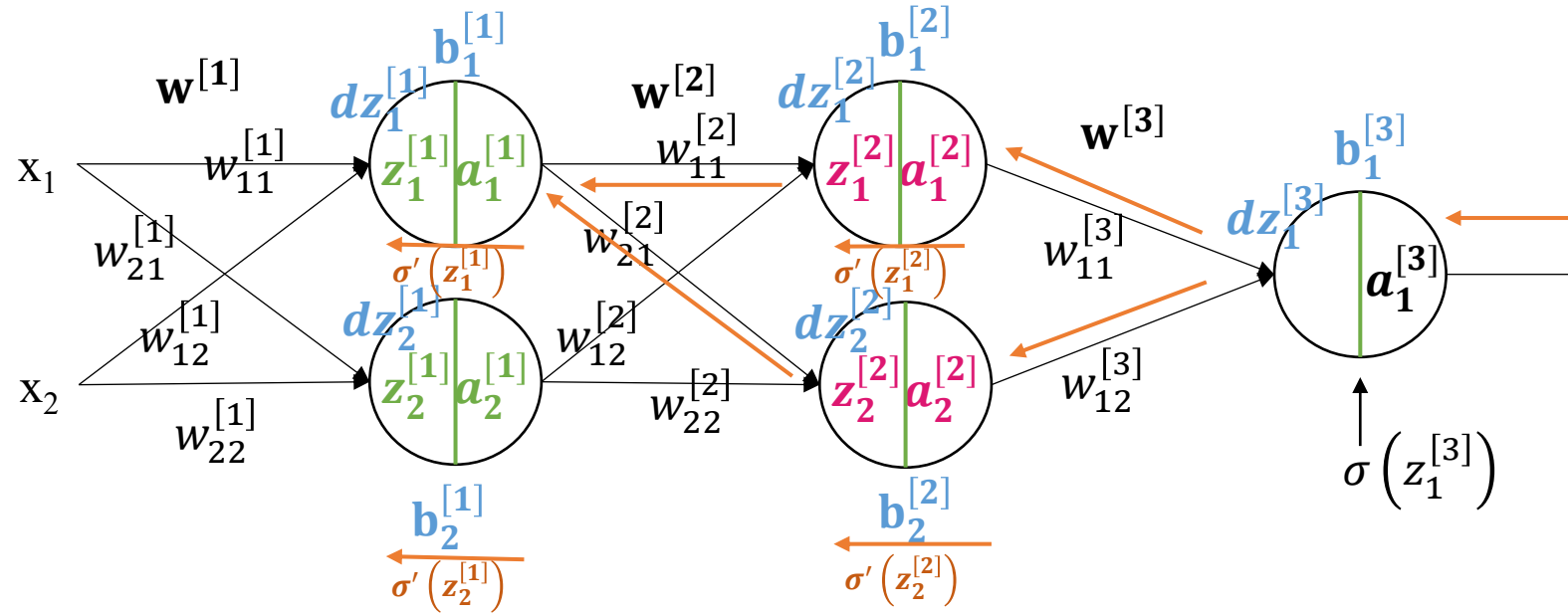
- Relu:

- $g(z) = \max(0, z) = \begin{cases} z & z \geq 0 \\ 0 & z < 0 \end{cases} \quad g'(z) = \begin{cases} 1 & \text{if } z \geq 0 \\ 0 & \text{if } z < 0 \end{cases}$

- Leaky Relu:

- $g(z) = \max(z, 0.01z) = \begin{cases} z & z \geq 0 \\ 0.01z & z < 0 \end{cases} \quad g'(z) = \begin{cases} 1 & \text{if } z \geq 0 \\ 0.01 & \text{if } z < 0 \end{cases}$

Back propagation(Bp)



$w_{ij}^{[l]}$

$$\frac{dL}{dw_{ij}^l} \quad \text{for } l = 1 \dots L$$

for each i,j

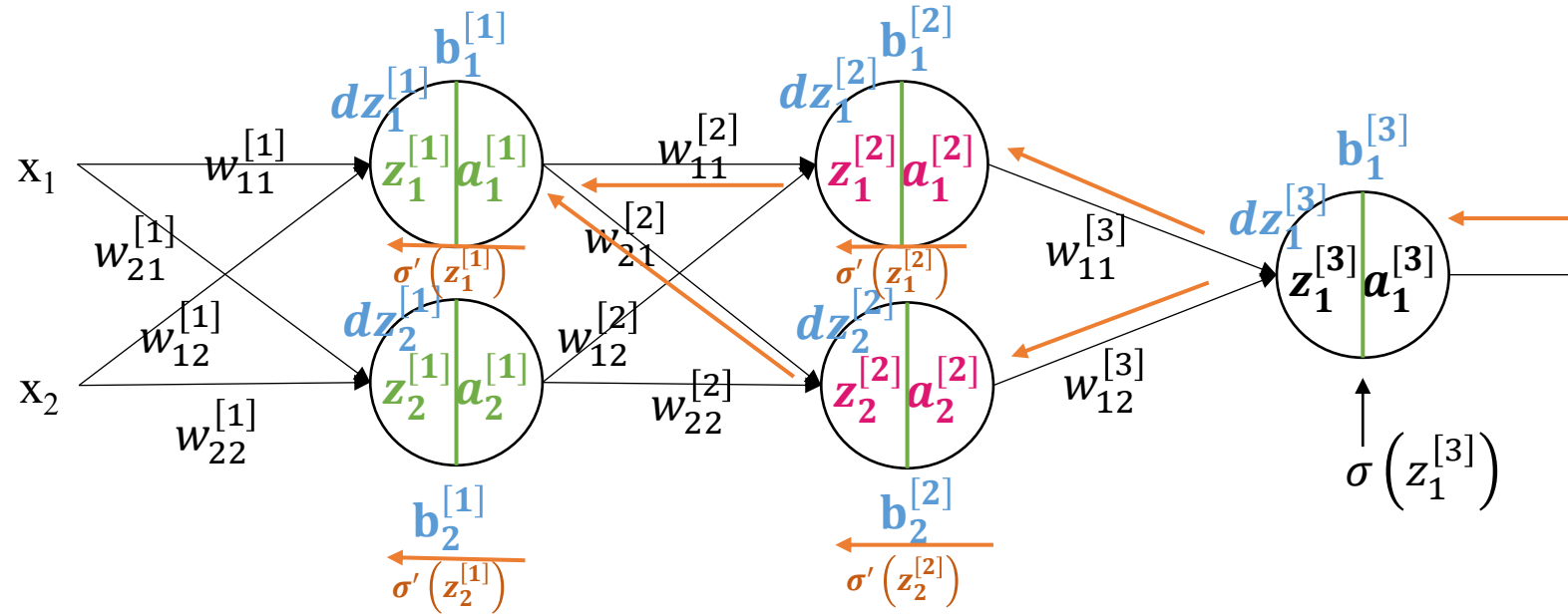
$$w_{ij}^{[l]} = w_{ij}^{[l]} - \alpha \frac{dL}{dw_{ij}^l}$$

$$\bullet dw_{11}^{[2]} = \frac{L(..., w_{11}^{[2]} + \varepsilon, ...) - L(..., w_{11}^{[2]} - \varepsilon, ...)}{2\varepsilon}$$

• پس انتشار

Back propagation(Bp)

• پس انتشار



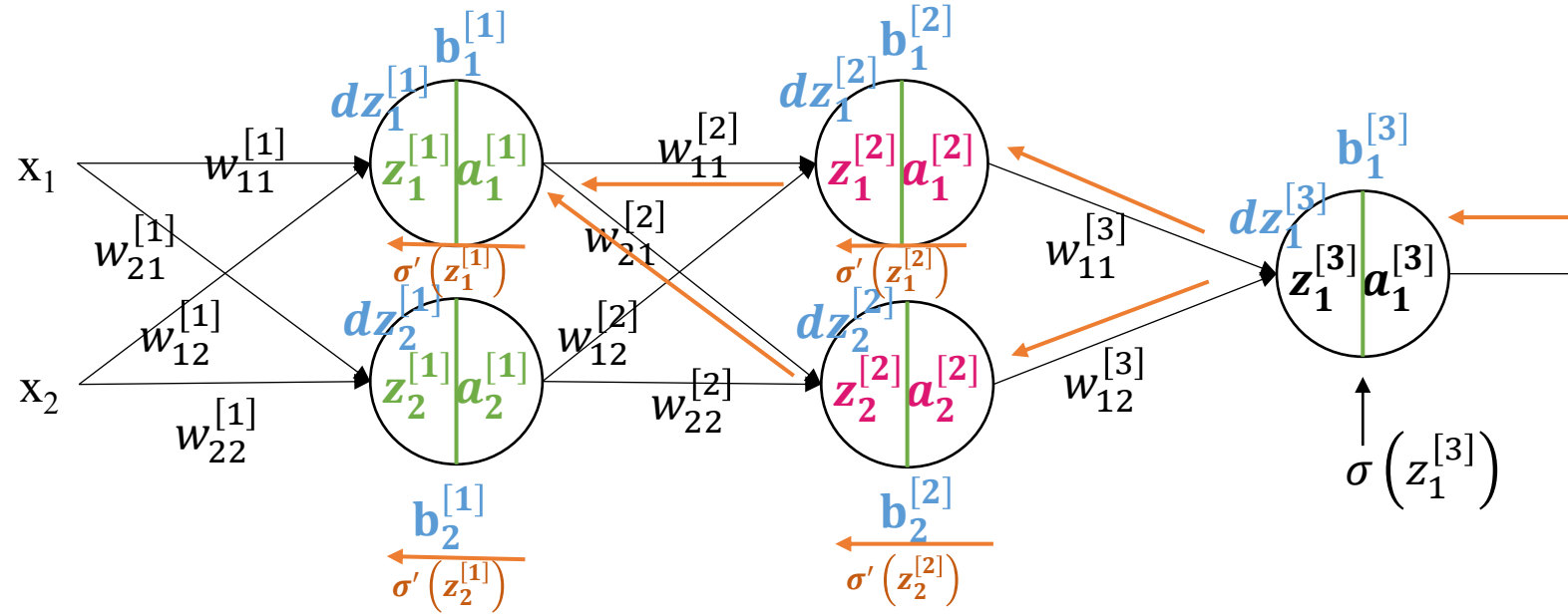
$$\frac{dL}{dw_{11}^{[3]}} = \underbrace{\frac{dL}{da_1^{[3]}} \times \frac{da_1^{[3]}}{dz_1^{[3]}}}_{\frac{dL}{dz_1^{[3]}} = dz_1^{[3]}} \times \frac{dz_1^{[3]}}{dw_{11}^{[3]}} = (a_1^{[3]} - y) a_1^{[2]}$$

$$z_1^{[3]} = a_1^{[2]} w_{11}^{[3]} + a_2^{[2]} w_{12}^{[3]} + b_1^{[3]}$$

$$\frac{dL}{db_1^{[3]}} = \frac{dL}{dz_1^{[3]}} \times \underbrace{\frac{dz_1^{[3]}}{db_1^{[3]}}}_1 = dz_1^{[3]} \times 1 = dz_1^{[3]}$$

Back propagation(Bp)

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$$dz_1^{[2]} = \frac{dL}{dz_1^{[2]}} = \underbrace{\frac{dL}{dz_1^{[3]}}}_{dz_1^{[3]}} \times \underbrace{\frac{dz_1^{[3]}}{da_1^{[2]}}}_{w_{11}^{[3]}} \times \underbrace{\frac{da_1^{[2]}}{dz_1^{[2]}}}_{\sigma'(z_1^{[2]})} \quad a_1^{[2]} = \sigma(z_1^{[2]}) \quad \frac{da_1^{[2]}}{dz_1^{[2]}} = \sigma'(z_1^{[2]})$$

$$dz_1^{[2]} = dz_1^{[3]} \times w_{11}^{[3]} \times \sigma'(z_1^{[2]})$$

$$dz_2^{[2]} = dz_1^{[3]} \times w_{12}^{[3]} \times \sigma'(z_2^{[2]})$$

$$dw_{11}^{[2]} = dz_1^{[2]} \times a_1^{[1]}$$

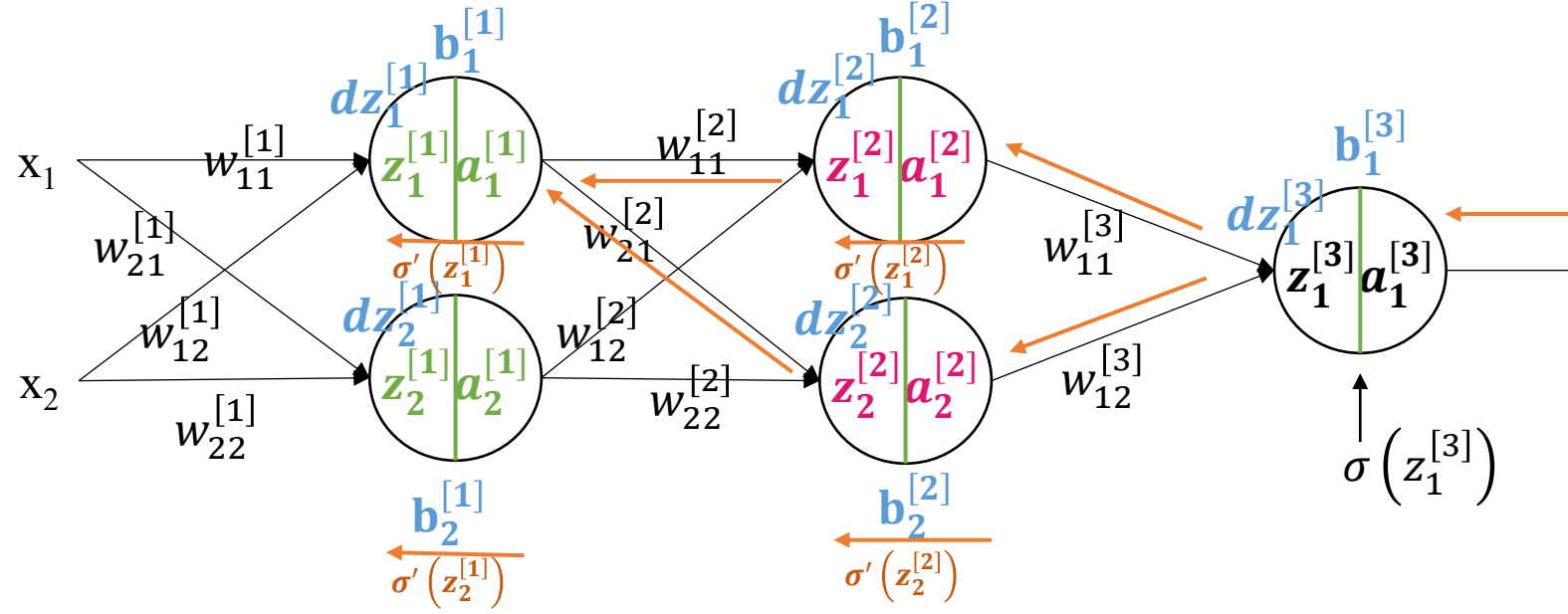
$$dw_{12}^{[2]} = dz_1^{[2]} \times a_2^{[1]}$$

$$db_1^{[2]} = dz_1^{[2]}$$

$$db_2^{[2]} = dz_2^{[2]}$$

Back propagation(Bp)

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$$dz_1^{[1]} = \frac{dL}{dz_1^{[1]}} = \underbrace{\frac{dL}{dz_1^{[2]}}}_{dz_1^{[2]}} \times \underbrace{\frac{dz_1^{[2]}}{da_1^{[1]}}}_{w_{11}^{[2]}} \times \underbrace{\frac{da_1^{[1]}}{dz_1^{[1]}}}_{\sigma'(z_1^{[1]})} + \underbrace{\frac{dL}{dz_2^{[2]}}}_{dz_2^{[2]}} \times \underbrace{\frac{dz_2^{[2]}}{da_1^{[1]}}}_{w_{21}^{[2]}} \times \underbrace{\frac{da_1^{[1]}}{dz_1^{[1]}}}_{\sigma'(z_1^{[1]})}$$

$$dz_1^{[1]} = (dz_1^{[2]} w_{11}^{[2]} + dz_2^{[2]} w_{21}^{[2]}) \sigma'(z_1^{[1]})$$

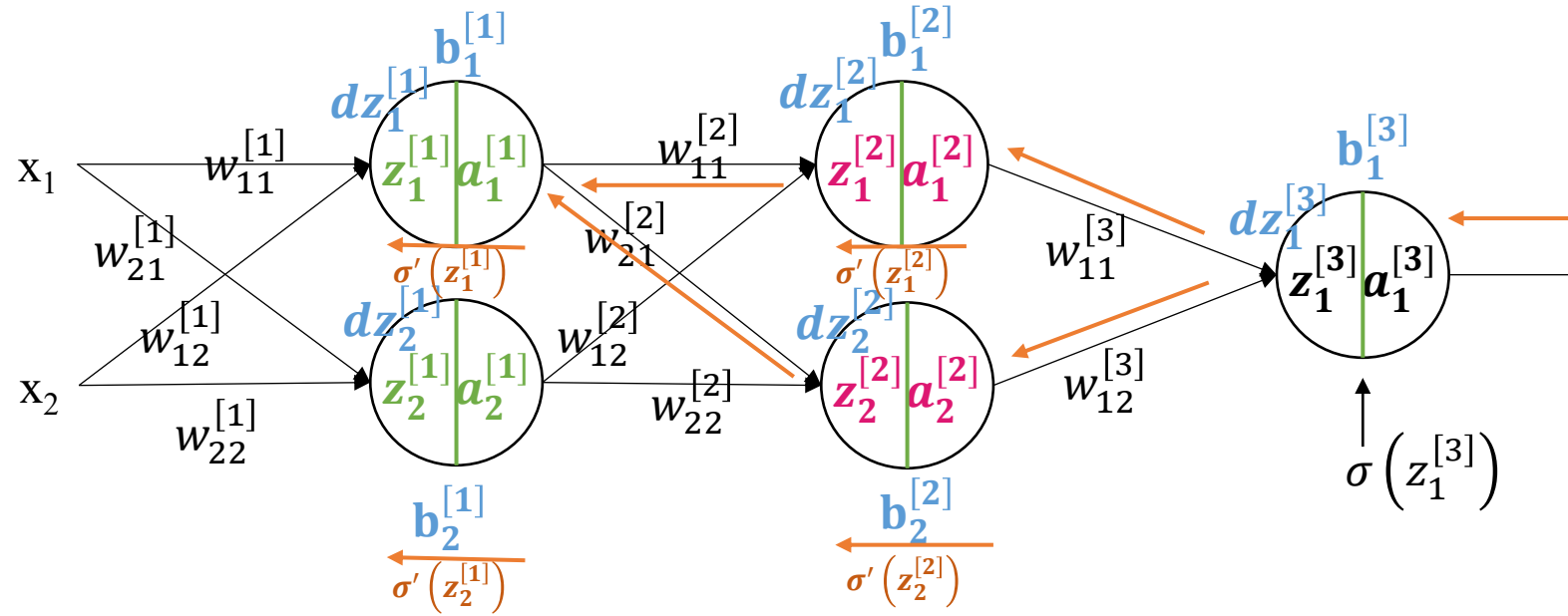
$$dz_2^{[1]} = (dz_1^{[2]} w_{12}^{[2]} + dz_2^{[2]} w_{22}^{[2]}) \sigma'(z_2^{[1]})$$

$$db_1^{[1]} = dz_1^{[1]}$$

$$db_2^{[1]} = dz_2^{[1]}$$

Back propagation(Bp)

• پس انتشار



$$dw_{11}^{[1]} = dz_1^{[1]} \times x_1$$

$$dw_{21}^{[1]} = dz_2^{[1]} \times x_1$$

$$dw_{12}^{[1]} = dz_1^{[1]} \times x_2$$

$$dw_{22}^{[1]} = dz_2^{[1]} \times x_2$$

Gradient descent

Gradient descent

- Parameters: $\underbrace{w^{[1]}}_{(n^{[1]}, n^{[0]})}$, $\underbrace{b^{[1]}}_{(n^{[1]}, 1)}$, $\underbrace{w^{[2]}}_{(n^{[2]}, n^{[1]})}$, $\underbrace{b^{[2]}}_{(n^{[2]}, 1)}$
- $w^{[1]}, b^{[1]}, w^{[2]}, b^{[2]} \leftarrow$ random initialization
- Repeat{
 - Forward propagation to compute $\hat{y}^{(i)}$ $i=1, \dots, m$
 - Backward prop
 - $dw^{[1]} = \frac{dJ}{dw^{[1]}}$, $db^{[1]}, dw^{[2]}, db^{[2]}$
 - Update
$$\begin{cases} w^{[1]} = w^{[1]} - \alpha dw^{[1]} \\ w^{[2]} = w^{[2]} - \alpha dw^{[2]} \\ b^{[1]} = b^{[1]} - \alpha db^{[1]} \\ b^{[2]} = b^{[2]} - \alpha db^{[2]} \end{cases}$$
 - }Until Convergence

Gradient descent

- *Forward propagation:*

$$z^{[1]} = w^{[1]}X + b^{[1]}$$

$$A^{[1]} = g^{[1]}(z^{[1]})$$

$$z^{[2]} = w^{[2]}A^{[1]} + b^{[2]}$$

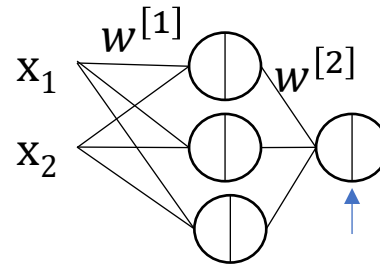
$$A^{[2]} = g^{[2]}(z^{[2]})$$

- *Back prop. :*

$$dz^{[2]} = A^{[2]}_{1 \times m} - Y_{1 \times m}$$

$$\underbrace{dw^{[2]}}_{1 \times n^{[1]}} = \frac{1}{m} \underbrace{dz^{[2]}}_{1 \times m} \underbrace{A^{[1]T}}_{m \times n^{[1]}}$$

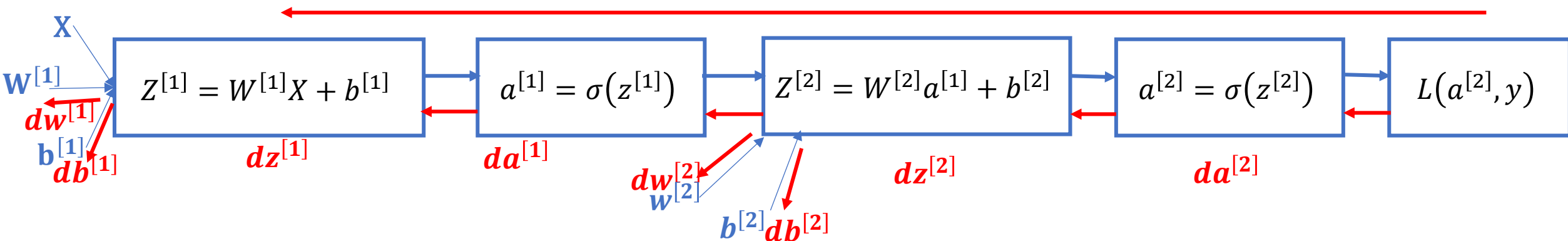
$$\underbrace{db^{[2]}}_{1 \times 1} = \frac{1}{m} np \cdot \text{sum} \left(\underbrace{dz^{[2]}}_{1 \times m}, axis = 1 \right) (1,1)$$



Gradient descent

- $$\underbrace{dz^{[1]}}_{n^{[1]} \times m} = \underbrace{w^{[2]T}}_{n^{[1]} \times 1} \underbrace{dz^{[2]}}_{1 \times m} * \underbrace{g^{[1]'}(z^{[1]})}_{n^{[1]} \times m}$$

Element wise
- $$\underbrace{dw^{[1]}}_{n^{[1]} \times n^{[0]}} = \frac{1}{m} \underbrace{dz^{[1]}}_{n^{[1]} \times m} \underbrace{X^T}_{m \times n^{[0]}}$$
- $$\underbrace{db^{[1]}}_{n^{[1]} \times 1} = \frac{1}{m} np \cdot \text{sum} \left(\underbrace{dz^{[1]}}_{n^{[1]} \times m}, \text{axis} = 1, \text{keepdims} = \text{True} \right)_{(n^{[1]}, 1)}$$



Gradient descent

- $dz^{[1]} = w^{[2]T} dz^{[2]} * g^{[1]'}(z^{[1]}) \quad m=1$
- $dz^{[2]} = a^{[2]} - y$
- $dw^{[1]} = dz^{[1]} X^T$
- $dw^{[2]} = dz^{[2]} a^{[1]T}$
- $db^{[1]} = dz^{[1]}$
- $db^{[2]} = dz^{[2]}$

Summary of gradient descent

- $dz^{[2]} = A^{[2]} - Y$
- $dW^{[2]} = \frac{1}{m} dz^{[2]} A^{[1]T}$
- $db^{[2]} = \frac{1}{m} np \cdot sum(dz^{[2]}, axis = 1, keepdims = True)$
- $dz^{[1]} = W^{[2]T} dz^{[2]} * g^{[1]'}(z^{[1]})$
- $dW^{[1]} = \frac{1}{m} dz^{[1]} X^T$
- $db^{[1]} = \frac{1}{m} np \cdot sum(dz^{[1]}, axis = 1, keepdims = True)$